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Examining spreadsheets and databases in relation to their use for fostering higher-order thinking skills in learners.

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This week we took a close look at spreadsheets and databases in relation to their use in developing higher-order thinking skills in learners. Please share with the class one idea in which you have had success with or would like to try with either a spreadsheet or database from the instructor perspective

I have done two postings on BB for this week's assignment, but for some weird reason they have disappeared. Anyway, I talked about some parts of this week's assignment last week. As I mentioned, database modeling requires many skills — data modeling, object-oriented modeling, system modeling, and so on. Database modeling is like a mind-modeling tool that groups things that are alike together (i.e., objects), and its methodology is the same as that used in other modeling tools. In data modeling, data is broken down into related data objects or tables. Each table has fields, data, and related fields (also known as relationships). An object's fields describe the property settings and data describes the behavior of an object.

In a nutshell, a database is a data structure that represents objects, data, and relationships. Most of my students understand what a relationship and a data set are, but they find it difficult to describe the meaning of an object and its relationship. An object is essentially a table. A row, or an instance of a table, describes an object and depends on its relationships and data. In software engineering, object-oriented programming is a programming paradigm that describes how objects are related to one another. Object-oriented modeling has the same semantics as database modeling. In database terms, a table is equivalent to an object, a field describes an object's property, and data is an atomic data value.

In contrast, a spreadsheet like Microsoft Excel is a data grid that organizes data as rows and columns. A single spreadsheet is equivalent to one table in the database. A table contains fields and data. The first row represents columns, and the second row and every row thereafter represent data.

I am teaching CS104B this semester at Ohlone College, and my class is currently building a "Bank Account Manager" application using the Microsoft .NET Framework. One of the objectives for the application is to design a database (data model). An example of a database contains the following tables: 1) BANK, 2) ACCOUNT, 3) ACCOUNT_TYPE, TRANSACTION, TRANSACTION_TYPE, and PERSON. Here is a URL to my class project:
http://www2.ohlone.edu/people/jpham/cs104b_class_project.html